Team Name :

Team Leader Name :

Team Size :

Problem Statement :

Submission Guidelines:
(Things to add on the submission deck)

1. Problem Brief

- What real business problem does this solve?
- Who is the target user/persona?
- What is the current pain point and how does this improve it?
- Industry/domain context

2. Architecture Diagram

- System design showing data flow
- Which CoCo CLI skills are used and how they connect
- Data sources (structured/unstructured)
- How modular components plug together

3. Impact Statement

- Measurable outcomes (time saved, accuracy improvement, etc.)
- Scalability potential
- How this extends beyond the demo

Governed Supply Chain Intelligence

Aerospace Manufacturing – Ontology-Driven Conversational Analytics

Problem Brief

The Business Problem

Different teams calculate the same metric differently. Planning reports OTD at 87%, Procurement at 91%, and Logistics at 84%. This metric chaos creates conflicting decisions, erodes trust in data, and misallocates resources.

How This Solves It

A governed ontology defines every KPI once. Dashboards, analytics, and AI responses use the same canonical formula, ensuring identical and trusted results across teams.

Industry Context

Aerospace supply chains operate under FAA compliance, long lead times, and supplier risk. Inconsistent metrics can delay production lines worth millions per day.

Target Users

Persona	Pain Point
VP of Supply Chain	Single trusted answer unavailable
Procurement Director	Metrics mismatch downstream reports
Operations Manager	Time spent reconciling numbers
Quality Director	Risk signals hidden in silos

ARCHITECTURE

Three-Tier Governed Data Flow with Conversational AI

STREAMLIT IN SNOWFLAKE (14 Pages)
Executive | Governance | Risk Intelligence | Conversational AI | Forecasting

SNOWFLAKE CORTEX COMPLETE
llama3.1-70b
Governed Prompt Context

GOVERNED SQL QUERIES
Cached 5-min TTL
Single Canonical Path

GOVERNED SQL QUERIES
Cached 5-min TTL
Single Canonical Path

ANALYTICS LAYER
V_METRIC_ALERTS (RAG status)
EXECUTIVE_KPI_MONTHLY
V_DERIVED_SUPPLIER_RISK
METRIC_THRESHOLDS (centralized)

RAW LAYER
6 Dimensions: Supplier, Part, Plant, Warehouse, Customer, Carrier
8 Facts: PO, Sales, Work Order, Shipment, Inventory, IoT, Quality

DATA FLOW: RAW (source of truth) --> ANALYTICS (governed logic) --> SEMANTIC (business meaning) --> AI + UI

Snowflake Skills: Cortex COMPLETE (LLM) | Semantic Views | Streamlit in Snowflake | Container Runtime | Governed Prompt Engineering

IMPACT STATEMENT

Measurable Outcomes, Scalability & Beyond the Demo

MEASURABLE OUTCOMES

Metric Reconciliation
4-6 hrs/wk --> 0 hrs = 100% eliminated

Cross-Team Agreement
~60% (3 definitions) = 100% consistent

Time-to-Insight
2-3 days (reports) = Real-time (AI) >95% faster

Risk Detection
Manual, weekly = Automated, continuous

Decision Confidence
Low (conflicting) = High (governed, auditable)

SCALABILITY

Horizontal: Add KPIs once, available everywhere

Vertical: Extend to Tier-2/3 suppliers

Cross-Enterprise: Replicable to pharma, auto, energy

AI Scaling: Swap LLM models without changing governance

BEYOND THE DEMO

Cortex Analyst for production NL queries

Dynamic Tables for sub-minute refresh

Snowflake Alerts on RAG threshold breaches

Marketplace sharing to supplier partners

Arctic-Extract for contract risk scoring

"The most expensive AI hallucination isn't a wrong answer - it's a plausible but ungoverned answer that different teams interpret differently. Ontology-first governance makes AI trustworthy at enterprise scale."

Built on: Snowflake | Cortex COMPLETE | Streamlit in Snowflake | Governed Semantic Views | DB: AEROSPACE_SUPPLY_CHAIN_AI

YOURSTORY | snowflake | H2S

CoCo CLI

HACKATHON

GCC Edition

THANK YOU